

AP Calc Lesson 2.6 Homework

Name _____

1. Which of the following expressions are equal to eight? Select all that apply.

A) $e^{\ln 8}$ E) $\ln e^4 + 2 \ln e^2$
 B) $e^{(\ln 4 + \ln 2)}$ F) $e^{\ln 2} \cdot e^{\ln 4}$
 C) $(\ln e^2)^3$ G) $\frac{4 \ln e^4}{\ln e^2}$
 D) $4 \ln e^2$ H) $\ln(e^2)^4$

2. $\frac{d}{dx}(2e^x - 5 \ln x + 4^x) =$

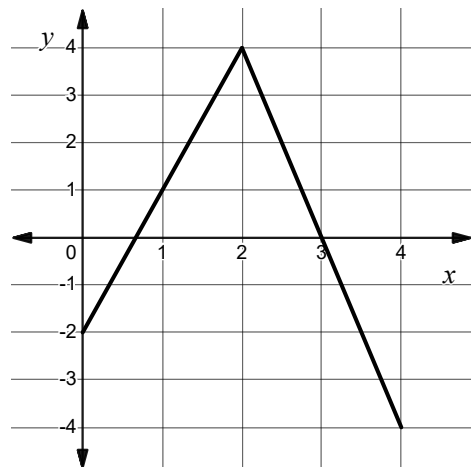
3. For what value of x does the line tangent to the graph of $y = 5e^x + 7x$ at $(0, 5)$ intersect the x -axis.

4. Let f be a differentiable function given by $f(x) = \begin{cases} 5e^x - ax & x < 0 \\ 3x + b \cos x & x \geq 0 \end{cases}$,
where a and b are constants. Find the values of a and b .

5. The graph of function f is shown in the figure. Let g be given by $g(x) = 5x - 6 \ln x$.

a. Find the value of x for which $g'(x) = f'(3)$.

b. If $h(x) = 2g(x) - f(x)$, find the equation of the line tangent to h at $x = 1$.



6. A line with slope m passes through the origin and is tangent to $y = 8 \ln x$. What is the value of m ?

7.

x	$f(x)$	$f'(x)$
0	-6	2
3	4	-1
5	9	7

The table gives values of a differentiable function f and its first derivative f' for selected values of x . Let g be the function given by $g(x) = -2x + 4 \ln x$. At what value of x is the instantaneous rate of change of g equal to the average rate of change of f on the interval $[0, 5]$?

8. Let f be the function given by $f(x) = \frac{e^x}{2} - x$. At what point on the graph of f is the tangent line parallel to $4x - 8y = 11$?
9. Let h be a positive constant and let $f(x) = \ln x$.
- a. Which is greater: $f(3)$ or $f(3 + h)$? How do you know?
- b. Which is greater: $f'(3)$ or $f'(3 + h)$? How do you know?

10. Let f be the function defined by $f(x) = \begin{cases} 7 \ln x & x \leq 1 \\ x^2 + 5x & x > 1 \end{cases}$. Which of the following statements is true?
- I. f is continuous at $x = 1$
 - II. $\lim_{x \rightarrow 1^-} f'(x) = \lim_{x \rightarrow 1^+} f'(x)$
 - III. $f'(1) = 7$
- A) I only
- B) II only
- C) II and III only
- D) I, II and III
11. Evaluate $\lim_{x \rightarrow e^3} \frac{\ln x - 3 \ln e}{x - e^3}$
- A) $\frac{3}{e}$
- B) $\frac{1}{e^3}$
- C) $\frac{1}{3}$
- D) The limit does not exist.